URBAN GOODS MOVEMENT

Broad Classes of Urban Goods Movement Demand

The classifications of urban goods based on spatial pattern of demand are broadly grouped as follows.

1. Goods movement between the urban areas and external locations

2. Inter industry goods movements within an urban area

3. Household based goods movements within an urban area

Terminology:

Unit: A unit receives certain type of commodity as input and dispatches other types of commodity as output.

A manufacturing plant: It receives the input of raw materials and semi furnished products and dispatches semi-finished and finished products for other destinations.

House hold: A house hold receives food and other consumer products and dispatch garbage for disposal.

Freight terminals: It receives goods that are either consolidated into large consignment for external locations are goods that must be separated into smaller consignments for distribution within the area.

External commodity Movements:

The commodity movements to from the external locations are termed as external commodity movements. It is of two types:

(a) Direct consignments: The major portions of direct consignments are made by trucks.

(b) Consignment via a freight terminal: Consignment via freight terminals involves pickup and delivery components by truck.

CLASSIFICATION OF URBAN GOODS MOVEMENT

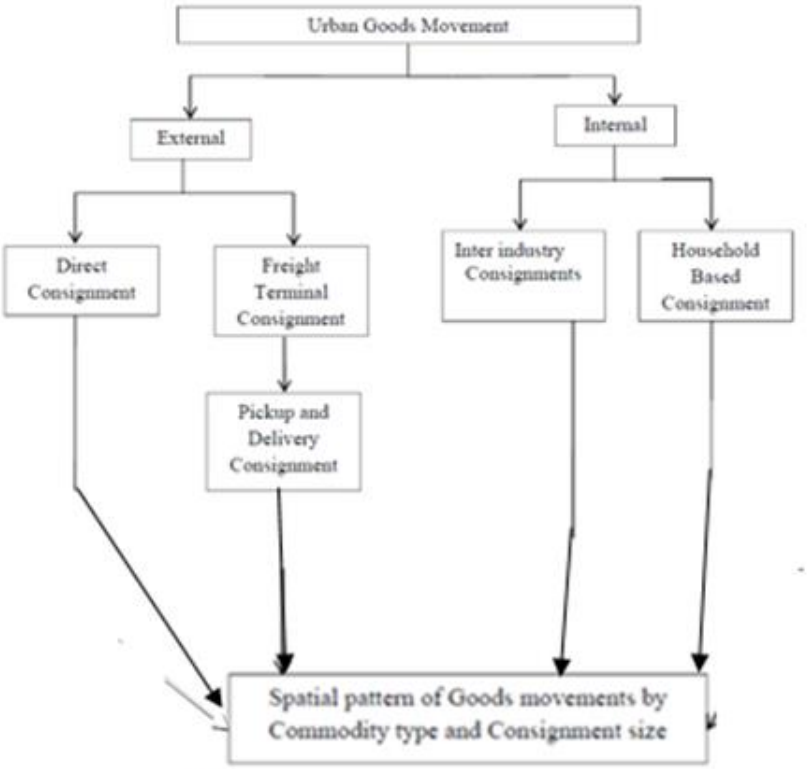

Figure 3.1: Classification of Urban Goods Movement

MODELLING DEMAND FOR URBAN GOODS TRANSPORT

(i) The freight transportation decision lies on economic optimization process i.e. to minimize the total cost associated with the movement of goods or with the economic activities involved in movement.

(ii) The freight transportation demand is totally derived from demand for the goods themselves at the point of consumption that are spatially separated from production point.

(iii) There is less fluctuation in the freight movement and thus peaking problem is eliminated.

Characteristic of the freight Transportation Demand:-

Level and Nature of freight activities:-

(1) **Quantity of Freight:**

Quantity affects the choice of mode -

- For bulk rail transport is preferred
- For limited quantity, road model are used

(2) **Type of Commodity:**

Certain Perishable commodities are required to be transported quickly than others; otherwise goods lose their utility and value.

(3) **Origin-Destination Specification:** It is required to design the transported networks and also for structural design of road pavements.

(4) **Trip length:**

Most of regional transportation demand has two types of traffic, long haul and short haul.

(5) **Level of service of modes:** Different modes will in general offer different levels of service and result in different costs for commodity transportation.

Characteristic of commodity transportation:-

(1) Total Travel time: This is a major attribute having influence over freight into line-haul time and terminal times at trip ends including loading and unloading as well as access to the line haul mode and delays.

(2) Total transport Cost: This is broken down into line haul freight rate, terminal cost including those for access and egress from the line haul mode, and terminal shortage cost at trip ends.

(3) Service Frequency: In freight transportation more frequent service can result in reduction in inventory levels and hence in storage costs for the shippers.

(4) System Reliability: System reliability means ability to predict the performance of the system in advance such as being able to anticipate the arrival of shipments and the state of goods being shipped. The reliability of the system generally reduces inventory levels and hence the freight costs.

(5) Preservability: For perishable goods decoration will occur at certain rate while in transit depending upon the manner in which the goods are packaged and carried hence needing quick transition.